

Recherche Opérationnelle

Programmation linéaire - Résolution graphique

Faire une résolution graphique.

Exercice 1

$$\left\{ \begin{array}{lll} \max & x_1 & + \quad 3x_2 \\ \text{s.c.} & x_1 & + \quad x_2 \leq 14 \\ & -2x_1 & + \quad 3x_2 \leq 12 \\ & 2x_1 & - \quad x_2 \leq 12 \\ & x_1, & x_2 \geq 0 \end{array} \right.$$

Exercice 2

$$\left\{ \begin{array}{lll} \min & x_2 & - \quad x_1 \\ \text{s.c.} & 2x_1 & - \quad x_2 \geq -2 \\ & x_1 & - \quad x_2 \leq 2 \\ & x_1 & + \quad x_2 \leq 5 \\ & x_1, & x_2 \geq 0 \end{array} \right.$$

Exercice 3

$$\left\{ \begin{array}{lll} \max & 2x_1 & + \quad x_2 \\ \text{s.c.} & x_1 & - \quad x_2 \leq 3 \\ & x_1 & + \quad 2x_2 \leq 6 \\ & -x_1 & + \quad 2x_2 \leq 2 \\ & x_1 & \geq 4 \quad x_2 \geq 0 \end{array} \right.$$

Exercice 4

$$\left\{ \begin{array}{lll} \max & x_1 & + \quad 2x_2 \\ \text{s.c.} & x_1 & - \quad \frac{1}{5}x_2 \leq 1 \\ & x_1 & + \quad x_2 \geq 6 \\ & -x_1 & + \quad x_2 = 3 \\ & x_1, & x_2, \geq 0 \end{array} \right.$$

Exercice 5

$$\left\{ \begin{array}{lll} \max(\min) & x_1 & + \quad 2x_2 \\ \text{s.c.} & -2x_1 & + \quad x_2 \leq 2 \\ & -x_1 & + \quad 2x_2 \leq 5 \\ & x_1 & - \quad 3x_2 \leq 4 \end{array} \right.$$